NETWORKING FUNDAMENTALS V


TEKsystems
Global Services
WORKFORCE
DEVELOPMENT

COURSE STRUCTURE

Day 1: Networking Fundamentals I

Day 2: Networking Fundamentals II

Day 3: Networking Fundamentals II

Day 4: Network Security in Depth

Day 5: Routing Fundamentals & Capstone

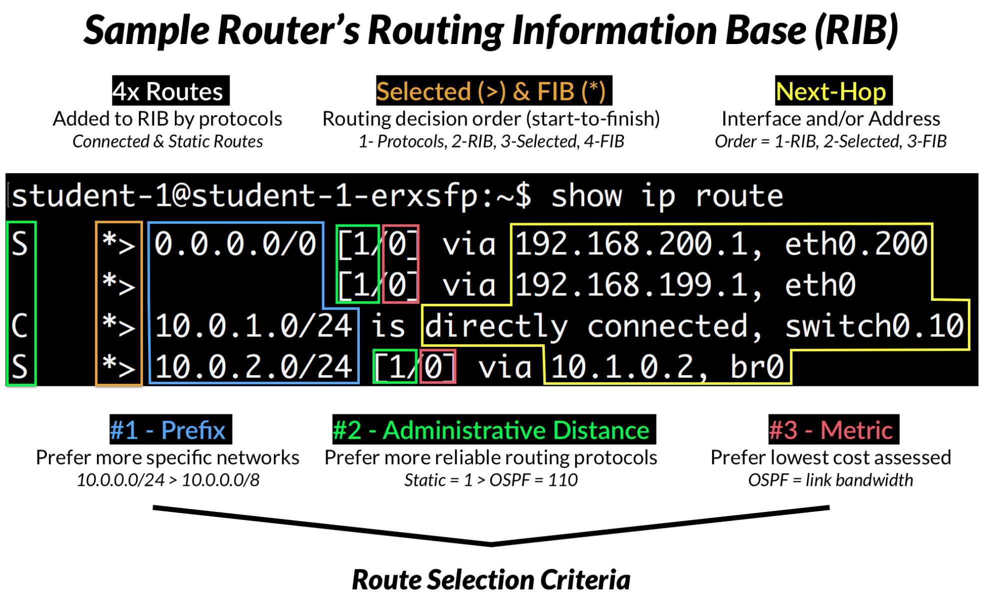
AGENDA FOR DAY 5

1. Static Routing: routes, AD, when to use static vs dynamic
2. Dynamic Routing: OSPF single-area, neighbour relationships, SPF
3. Reading the Routing Table — codes, longest match, AD tie-breaking
4. Knowledge Checks
5. Capstone Lab Review — Days 1–4 summary
6. Lab 09: Multi-Router Static Routing
7. Lab 10: OSPF Single-Area Configuration
8. Lab 11: Full Network Capstone

STATIC & DYNAMIC ROUTING

HOW ROUTERS MAKE FORWARDING DECISIONS

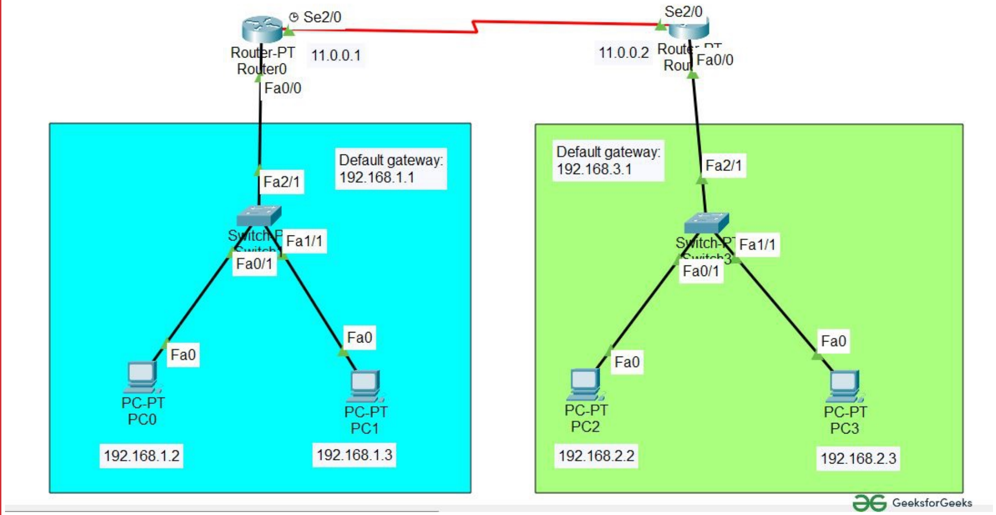
- **Routing Table:** Every router maintains a routing table. A list of known networks and how to reach them
- **Routing table sources:** Connected (C), Static (S), OSPF (O), EIGRP (D), RIP (R) — code shown in show ip route
- **Administrative Distance (AD):** when two sources know the same prefix, lowest AD wins — Connected=0, Static=1, OSPF=110
- **Metric:** when same protocol knows multiple paths to same prefix, lowest metric wins
- **Longest prefix match:** /28 beats /24 beats /0 for a matching destination — most specific route always wins
- **Default route (0.0.0.0/0):** gateway of last resort — matches everything with no more-specific route
- **show ip route:** full routing table | `show ip route 10.0.1.5` — which route matches this destination



RIB (Routing Information Base)** — collects every route from every source; AD determines which entry wins and gets installed into the routing table

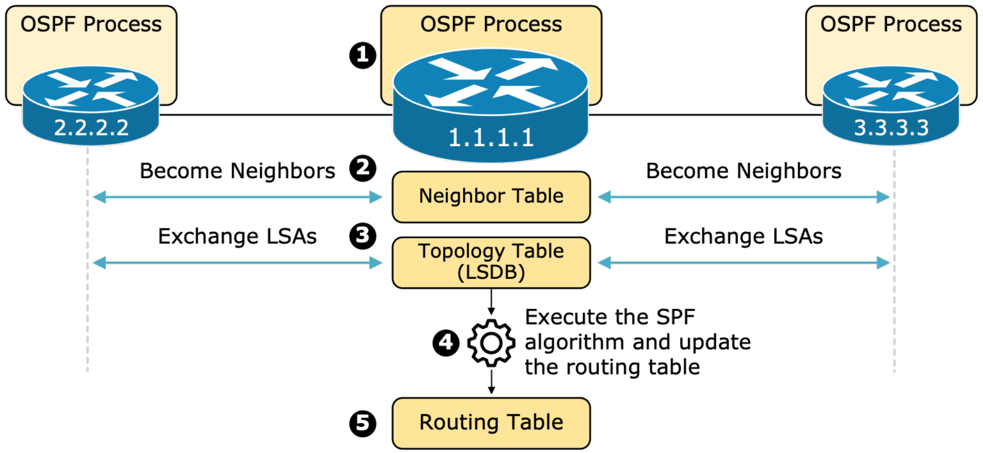
STATIC ROUTING

- **Static route:** manually configured path to a destination — AD=1, never changes unless you change it
- **ip route 10.0.2.0 255.255.255.0 10.0.1.2 — to reach 10.0.2.0/24, send packets to next-hop 10.0.1.2**
- **ip route 10.0.2.0 255.255.255.0 Gi0/1 — alternative: exit interface instead of next-hop**
- **Default static route: ip route 0.0.0.0 0.0.0.0 203.0.113.1 —** use this router as internet gateway
- **Recursive lookup:** router looks up the next-hop IP in the routing table to find the exit interface
- **Floating static route:** ip route 10.0.2.0 255.255.255.0 10.0.1.3 200 — AD=200, only used if OSPF route disappears
- **Use static routes for:** stub networks, default routes, specific overrides — not for large dynamic networks
- **Verify:** `show ip route static` | `show ip route 10.0.2.0`



OSPF — OPEN SHORTEST PATH FIRST

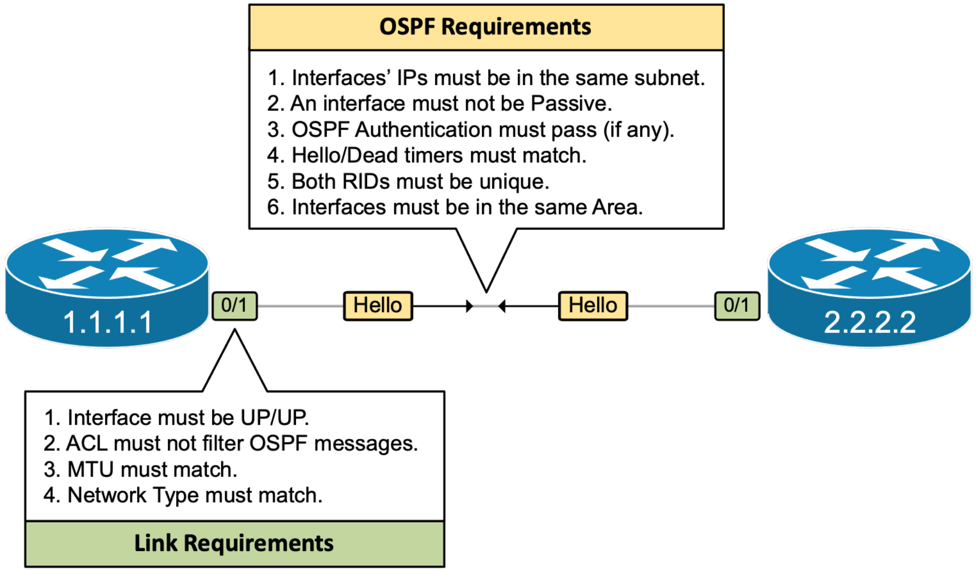
- **Link-state protocol:** every router builds a complete map of the network (LSDB) and runs SPF algorithm locally
- **All routers in an area share identical LSDB:** LSAs (Link State Advertisements) flood between all routers
- **SPF (Dijkstra) algorithm:** each router independently calculates the shortest path to every destination
- **Metric = cost = reference bandwidth** / interface bandwidth; 10Gbps=1, 1Gbps=1, 100Mbps=1 (adjust ref BW!)
- **router ospf 1:** start OSPF process | network 10.0.0.0 0.0.0.255 area 0 — advertise network
- **Wildcard mask is inverse of subnet mask:** /24 = 0.0.0.255 | /30 = 0.0.0.3 | /16 = 0.255.255.255
- **AD = 110:** preferred over RIP (120) but not static (1) or connected (0)
- `show ip ospf neighbor | show ip ospf database | show ip route ospf`



LSDB (Link State Database) — shared map of every router and link in the area; identical on every OSPF router; SPF runs against it locally to calculate best paths

OSPF NEIGHBOR RELATIONSHIPS

- **Neighbors must agree on:** area ID, hello/dead timers, subnet, authentication, MTU — mismatch = stuck in INIT
- **Hello packets:** sent every 10s on point-to-point, 10s on broadcast; dead interval = 4x hello (40s default)
- **DR/BDR election on multi-access networks (Ethernet):** highest priority wins (default 1), tie-break = highest router ID
- **Router ID:** highest loopback IP, else highest physical interface IP — set manually: router-id 1.1.1.1
- **Neighbor states: Down → Init → 2-Way → ExStart → Exchange → Loading → Full**
- **Full adjacency required before routes are shared:** stuck in 2-Way or ExStart = MTU or duplex issue
- **passive-interface Gi0/2:** advertise network but suppress hellos (use on LAN-facing ports with no OSPF neighbors)
- `show ip ospf neighbor` — check state is Full **|** `show ip ospf interface` — hello/dead timers, DR/BDR



READING THE ROUTING TABLE

- C = Connected (AD 0) — directly attached network; L = Local (AD 0) — router's own interface IP (/32)
- S = Static (AD 1) | O = OSPF (AD 110) | D = EIGRP (AD 90) | R = RIP (AD 120) | * = candidate default route
- **Format:** O 10.0.2.0/24 [110/2] via 10.0.1.2, 00:01:23, GigabitEthernet0/1
-   [110/2] = [AD/metric] — 110 is OSPF AD, 2 is the cumulative cost
-   via 10.0.1.2 = next-hop IP | GigabitEthernet0/1 = exit interface
- **Longest match wins:** packet to 10.0.1.5 — /30 beats /24 beats /0 — router picks most specific
- **If no match and no default route:** packet dropped, ICMP unreachable sent back to source
- **show ip route 10.0.1.5:** shows exactly which route entry would match that destination address

POP QUIZ:

A router has two routes to 10.0.5.0: a static route (AD=1) and an OSPF route (AD=110). Which does it install in the routing table?

A. OSPF route — it was learned dynamically
B. Static route — lower AD wins
C. Both are installed and load-balanced
D. The one with the lower metric

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POP QUIZ:

Two OSPF routers are stuck in the 2-Way state and never reach Full adjacency. What is the most likely cause?

A. Mismatched OSPF area IDs
B. Different router IDs configured
C. MTU mismatch between the two interfaces
D. One router has a higher OSPF priority

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POP QUIZ:

A routing table has: S 10.0.0.0/8, O 10.0.1.0/24, C 10.0.1.0/30. A packet arrives for 10.0.1.2. Which route is used?

A. S 10.0.0.0/8 — lowest AD (static)
B. O 10.0.1.0/24 — OSPF is most reliable
C. C 10.0.1.0/30 — longest prefix match wins
D. Load-balanced across all three

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C. **C 10.0.1.0/30 — longest prefix match wins**
D. Load-balanced across all three

COURSE REVIEW & CAPSTONE

DAYS 1 & 2 RECAP — FOUNDATIONS & ADDRESSING

- **Networks connect devices via endpoints**, links, and intermediaries — scales from home to global internet
- **OSI 7-layer model**: physical → data link → network → transport → session → presentation → application
- **TCP** = reliable, connection-oriented (SYN/SYN-ACK/ACK); **UDP** = fast, connectionless
- **IPv4**: 32-bit dotted decimal; RFC1918 private ranges; CIDR /N notation; subnetting = block size 256−mask
- **IPv6**: 128-bit hex; link-local fe80::/10; auto-assigned /64; no broadcast
- **DNS**: resolves names → IPs; A/AAAA/CNAME/MX/PTR records; recursive vs authoritative
- **DHCP**: Discover → Offer → Request → Ack; relay agent for inter-subnet DHCP
- **NAT/PAT**: RFC1918 private IPs share one public IP; PAT uses port numbers to track sessions

DAY 3 RECAP — TROUBLESHOOTING & VERIFICATION

- **Structured troubleshooting:** define → gather → hypothesise → test → eliminate → document
- **Top-down (L7→L1) vs bottom-up (L1→L7):** start bottom-up when no clue; top-down when app-specific
- **ping 127.0.0.1 =** local stack | ping gateway = L1–L3 | ping remote = routing | tracert = path mapping
- **show ip interface brief:** status/protocol per interface | show ip route — routing table
- **show arp** / arp -a — IP→MAC mappings | show mac address-table — switch CAM table
- **show interfaces:** error counters (CRC, runts, giants) reveal physical layer problems
- **Wireshark:** capture and decode frames in real time; filter by protocol, IP, port
- **Pipe modifiers:** show run | include ip address | show ip route | section interface

DAY 4 RECAP — NETWORK SECURITY

- **CIA Triad:** Confidentiality (encryption), Integrity (hashing), Availability (redundancy)
- **L2 attacks:** MAC flooding, ARP spoofing, VLAN hopping, STP manipulation
- **L2 defences:** Port Security (max MACs), DHCP Snooping (trusted ports), DAI (validates ARP vs binding table)
- **BPDU Guard:** err-disables port if BPDU received on PortFast port — stops rogue switch attacks
- **Firewalls:** stateless (per-packet) → stateful (connection tracking) → NGFW (app + IDS/IPS + URL)
- **DMZ:** public-facing servers isolated between two firewall zones — compromise doesn't reach inside
- **ACLs:** standard (source IP only, near dest) vs extended (src+dst+port+proto, near source); implicit deny at end
- **Zero Trust:** never trust, always verify — identity is the new perimeter; micro-segmentation east-west

DAY 5 RECAP — ROUTING

- **Routing table decision order:** longest prefix match first → then lowest AD → then lowest metric
- **Connected (0) → Static (1) → OSPF (110) → RIP (120) —** lower AD always wins for same prefix
- **Static route:** ip route dest mask next-hop — simple, manual, best for stub networks and default routes
- **Floating static:** higher AD than dynamic route — backup path that only activates on primary failure
- **OSPF:** link-state, all routers share LSDB, SPF runs locally, metric = cost (ref BW / interface BW)
- **OSPF neighbours must match:** area, hello/dead timers, subnet, authentication, MTU
- **Full adjacency =** routes shared; stuck in 2-Way/ExStart = MTU or timer mismatch
- **passive-interface** suppresses hellos on end-device ports while still advertising the network

LAB 12: MULTI-ROUTER STATIC ROUTING

- Goal: Build a three-router network and configure static routes so all subnets can communicate
- Steps:
  - Build topology: R1–R2 on 10.0.12.0/30 | R2–R3 on 10.0.23.0/30 | R1 LAN 192.168.1.0/24 | R3 LAN 192.168.3.0/24
  - Assign IPs on all interfaces and bring them up; add a PC to R1 and R3 LANs with correct gateways
  - R1 static routes: ip route 10.0.23.0 255.255.255.252 10.0.12.2 | ip route 192.168.3.0 255.255.255.0 10.0.12.2
  - R2 static routes: ip route 192.168.1.0 255.255.255.0 10.0.12.1 | ip route 192.168.3.0 255.255.255.0 10.0.23.2
  - R3 static routes: ip route 10.0.12.0 255.255.255.252 10.0.23.1 | ip route 192.168.1.0 255.255.255.0 10.0.23.1
  - Test: PC on R1 LAN pings PC on R3 LAN | tracert confirms R2 is hop 2
  - Verify: show ip route on each router | remove one route and observe ping failure

LAB 13: OSPF SINGLE-AREA CONFIGURATION

- Goal: Replace all static routes with OSPF and verify automatic route learning between three routers
- Steps:
  - Use the same three-router topology from Lab 09 — remove all static routes: no ip route 0.0.0.0 etc.
  - Configure OSPF on R1: router ospf 1 / network 192.168.1.0 0.0.0.255 area 0 / network 10.0.12.0 0.0.0.3 area 0
  - Configure OSPF on R2: router ospf 1 / network 10.0.12.0 0.0.0.3 area 0 / network 10.0.23.0 0.0.0.3 area 0
  - Configure OSPF on R3: router ospf 1 / network 10.0.23.0 0.0.0.3 area 0 / network 192.168.3.0 0.0.0.255 area 0
  - Suppress hellos on LAN interfaces: passive-interface Gi0/2 on R1 and R3
  - Verify neighbours: show ip ospf neighbor — confirm Full state on all router-to-router links
  - Verify routes: show ip route ospf — confirm O routes learned; compare to Lab 09 static routes
  - Test failover: shut one link, observe OSPF reconverge and re-route; bring link back up

LAB 14: FULL NETWORK CAPSTONE

- Goal: Build a complete network from scratch applying all skills from Days 1–5 without step-by-step guidance
- Steps:
  - Design: Site A has HR (VLAN 10) and Guest (VLAN 20); Site B has a server subnet; sites connected via router-to-router link
  - Site A switch: VLANs 10 and 20, trunk to router, access ports for PCs, native VLAN 99
  - Router-on-a-stick: sub-interfaces Gi0/0.10 and Gi0/0.20 for inter-VLAN routing at Site A
  - OSPF: advertise all subnets into area 0 on both routers; verify Full adjacency
  - DHCP: configure pools on Site A router for VLAN 10 and 20; verify PCs receive addresses
  - ACL: deny VLAN 20 from reaching Site B server subnet; permit all else; apply near source
  - L2 security: DHCP Snooping and DAI on VLANs 10 and 20; port security max 1 on access ports
  - Verify: VLAN 10 reaches Site B; VLAN 20 cannot | show ip route, show ip ospf neighbor, show access-lists

WHAT YOU'VE BUILT — AND WHAT'S NEXT

- You now understand how data moves across any network. From bits on a wire to application data
- You can build, configure, secure, and troubleshoot a multi-subnet network from scratch
- These fundamentals are the foundation every network engineer builds on every day
- Routing & Switching Essentials (next week) goes deeper on:
- VLANs, trunking, STP, EtherChannel, HSRP: enterprise switching in depth
- OSPF multi-area, EIGRP, BGP overview, route redistribution
- Data center architecture, cloud networking, SD-WAN
- Tools you now know:
- Cisco IOS CLI | Packet Tracer | Wireshark | ping / tracert / show commands